Transfer Attacks in Face Recognition

DPA-HMA (Adapted) - Diverse Parameters Augmentation with Hard-view Momentum Attack

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21 June, 2026

Reference Paper

"Improving the Transferability of Adversarial Attacks on Face Recognition with Diverse Parameters Augmentation" - CVPR 2025

Executive Summary

DPA-HMA (Adapted) achieves the highest breach rate among all evaluated vanilla transfer attacks, demonstrating the effectiveness of diverse-view gradient aggregation for cross-model adversarial transferability in face recognition.

39.58%

Overall Breach Rate

190 of 480 evaluation cases successfully breached

0.214

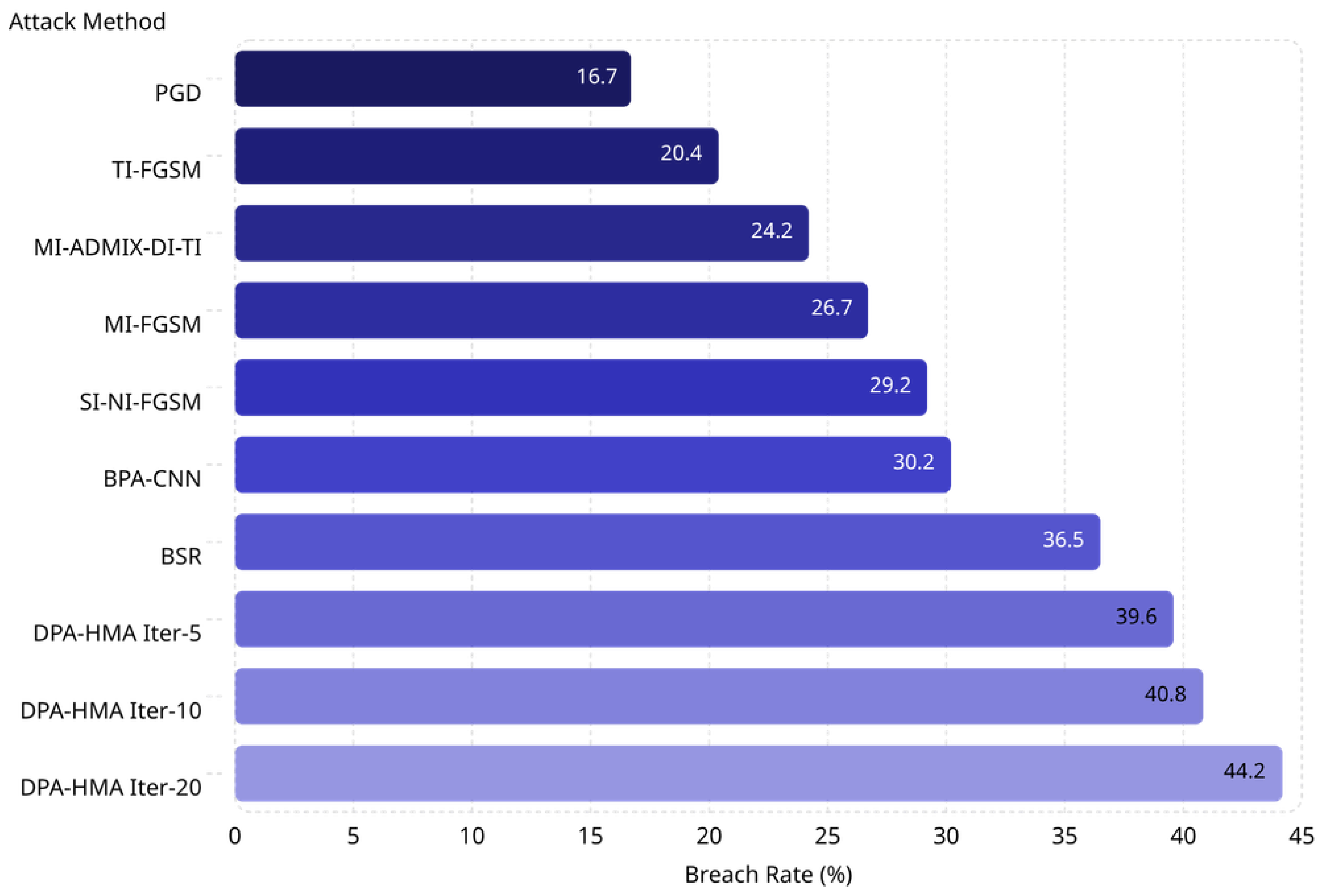
Mean Embedding Impact

Cosine distance shift across all victim models

+10.4pp

Gain over Best Vanilla

Over SI-NI-FGSM (29.2%) - highest vanilla baseline



Rank #1 Vanilla

+3.1 pp over prior best student attack (BSR: 36.5%)

+15 Extra Breaches

190 vs 175 successful breaches compared to BSR

No Impact Trade-off

Mean impact 0.214 vs BSR 0.205 - marginal difference

DPA-HMA (Adapted): Method Overview

CVPR 2025 | WHITE-BOX, GRADIENT-BASED, L∞-BOUNDED

Defintion & Nature

DPA-HMA combines **Diverse Parameters Augmentation** (DPA) with **Hard-view Momentum Attack** (HMA) - a CVPR 2025 transfer attack adapted to the single-surrogate DeepFace pipeline.

- Input-transformation attack with hard-view gradient aggregation
- Momentum PGD with sign-gradient updates; $\varepsilon = 0.062$ (L_∞), $T = 5$ iterations
- One surrogate DeepFace model per run; evaluated on unseen victim models

Goal in Face Recognition

- **Impersonation:** maximize cos(adv, target) - loss = mean(cos)
- **Dodging:** minimize cos(adv, target) - loss = mean(1 − cos)
- Success = threshold crossing on victim model (dataset-specific thresholds)
- Cross-model transfer: perturbation from surrogate A tested on victim B (B ≠ A)

Core Mechanism

Problem: Single-view PGD/MI-FGSM gradients overfit the surrogate model, reducing transferability to victim architectures.
DPA-HMA builds **8 diverse copies per iteration**:

- **Hard nudge:** adv + step × sign(last_grad), steps ∈ {0, ±0.004, ±0.008}
- **Random affine warp:** scale + rotation + shear noise ∈ [0.01, 0.08]
- Average loss over 8 copies → smoother, more transferable gradient

Pseudocode

```
Input: source x, target embedding e_t, model f
       ε=0.062, T=5; adv←x; g←0; hard_grad←0; α←ε/T

for t = 1…T:
  X ← empty batch
  for i = 1…8:
    x̃ ← clip(adv + step_i × sign(hard_grad))
    X[i] ← AffineWarp(x̃)   // random projective transform
  L ← AttackLoss(f(X), e_t) // cos or 1-cos
  grad ← ∇_adv L ; hard_grad ← grad
  g ← g + normalize(grad)    // momentum, decay=1.0
  adv ← adv + α × sign(g)
  adv ← Proj_ε(adv, x) ; clip(adv, -1, 1)
return adv
```

Evaluation Environment

EXPERIMENTAL SETUP

The evaluation is designed to rigorously assess cross-model adversarial transferability under a shared, constrained perturbation budget across multiple face recognition architectures and attack goals.

1 **Scale**

30 face pairs × 2 goals × 4 attackers × 4 non-self victims = **480 evaluation cases**

2 **Attacker Models**

Facenet512, ArcFace, GhostFaceNet, VGG-Face - all via DeepFace framework

3 **Victim Models**

Facenet512, ArcFace, GhostFaceNet, VGG-Face, IR152 - self-transfer excluded from evaluation

4 **Attack Goals**

Impersonation (240 cases): adversarial image should match target identity; **Dodging** (240 cases): adversarial image should evade own identity

Shared Perturbation Budget

Parameter	Value
ε (L_∞ norm bound)	0.062
Iterations (T)	5
Momentum decay	1.0
Step size (α)	$\varepsilon / T = 0.0124$

Evaluation Metrics

Breach Rate

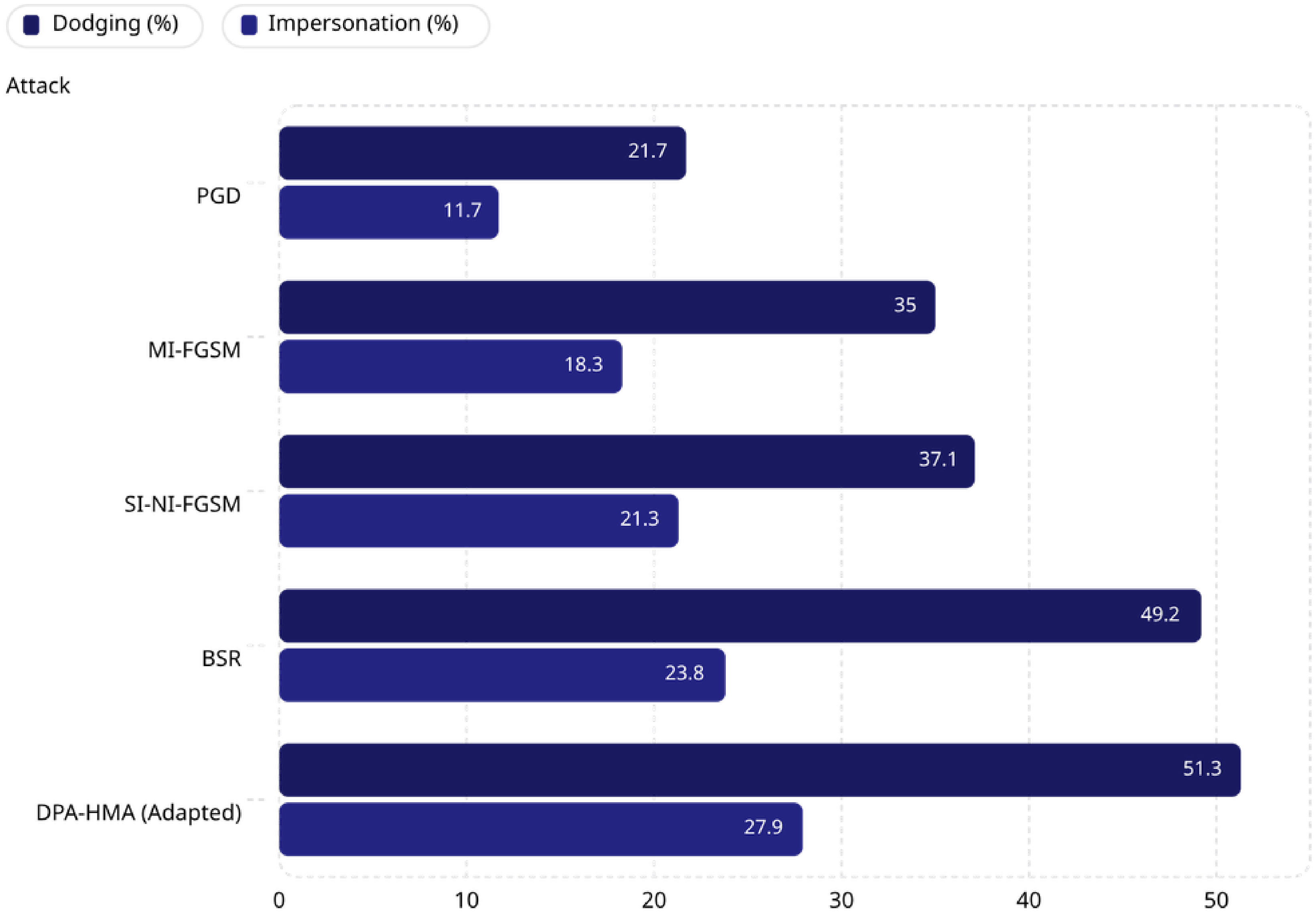
Percentage of cases where adversarial image crosses victim model's threshold (%)

Mean Embedding Impact

Average cosine distance shift in victim embedding space induced by the perturbation

Performance by Attack Goal

Decomposing the 480 evaluation cases by goal reveals that DPA-HMA (Adapted) achieves its primary advantage in impersonation - the harder and more security-critical task - while matching the best prior student attack on dodging.



Impact Breakdown (Mean Cosine Shift)

Attack	Dodging	Impersonation
DPA-HMA (Adapted)	0.276	0.151
BSR	0.266	0.143
SI-NI-FGSM	0.223	0.128

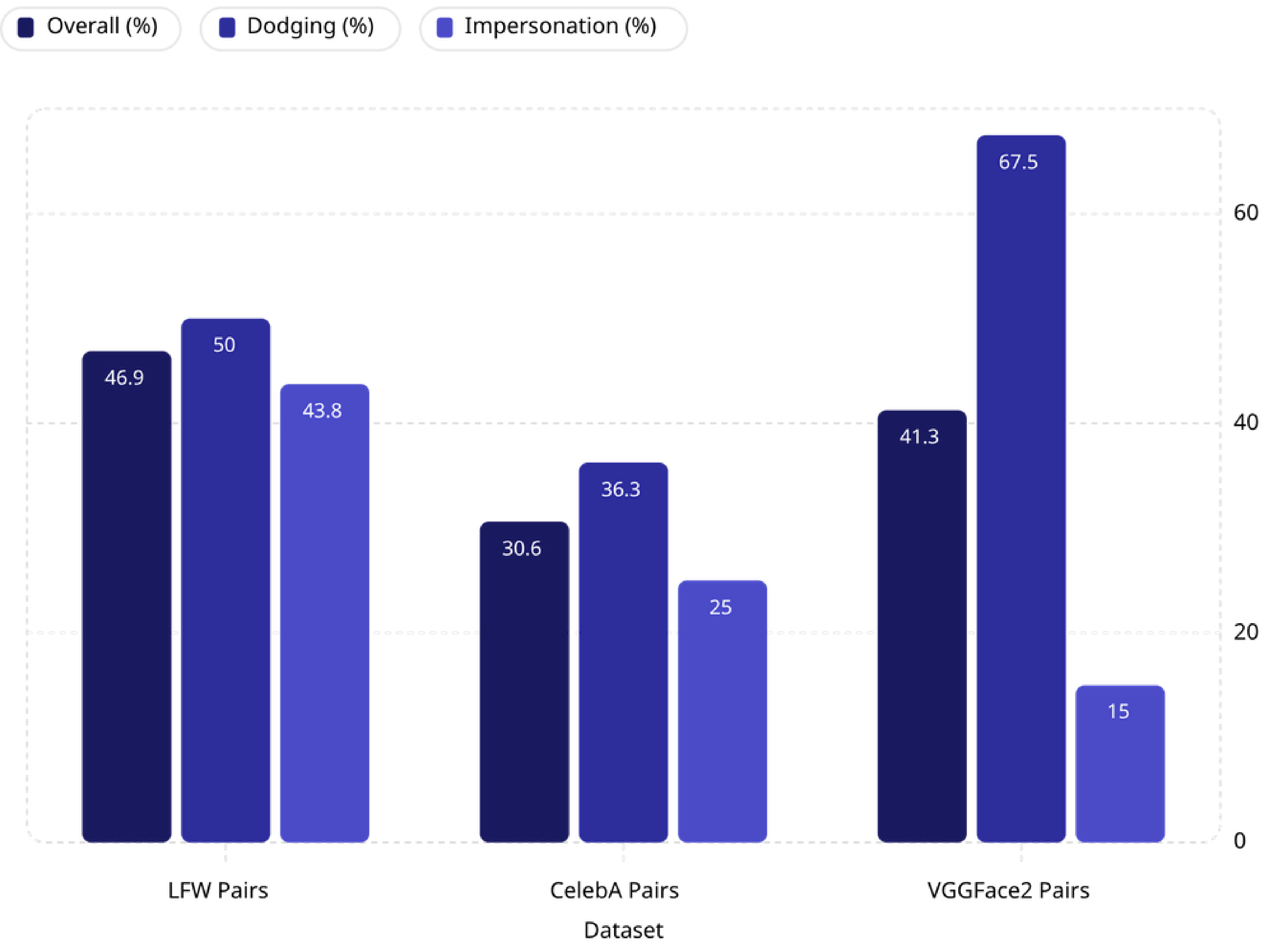
Dodging: DPA-HMA outperforms BSR with **51.25% vs 49.17%**, and is **+14.17 pp** over SI-NI-FGSM at 37.08%.

Impersonation: DPA-HMA reaches **27.92%,** the best among compared attacks, with **+4.17 pp over BSR** and **+6.67 pp over SI-NI-FGSM**.

The gain is driven by improvements in both dodging and impersonation, with the strongest relative advantage appearing in impersonation. This indicates that diverse-view gradient averaging improves cross-identity attack generalization while also improving identity-evasion transfer.

Performance Across Face Datasets

DPA-HMA was evaluated across three benchmark face pair datasets LFW, CelebA, and VGGFace2 each contributing 160 of the 480 total cases (80 dodging + 80 impersonation).



Dataset	Overall	Mean Impact
LFW Pairs	46.88%	0.225
CelebA Pairs	30.63%	0.235
VGGFace2 Pairs	41.25%	0.181

VGGFace2 Dodging: 67.50% - highest single dataset-goal score recorded across all experiments. Identity diversity in VGGFace2 may facilitate evasion.

VGGFace2 Impersonation: 15.0% - hardest setting. Cross-identity matching across a large-scale, high-diversity dataset proves most challenging for transfer attacks.

LFW shows the most balanced profile (46.88% overall) - nearest to real-world constrained verification. **CelebA** yields the lowest overall breach (30.63%), possibly due to varied image conditions.

Cross-Model Transferability

Adversarial images generated on row (attacker) → evaluated on column (victim) | 30 pairs each | diagonal excluded

Breach Rate (%)

Attacker ↓ / Victim →	Facenet512	ArcFace	GhostFaceNet	IR152	VGG-Face
Facenet512	-	60.0	43.3	46.7	63.3
ArcFace	36.7	-	43.3	26.7	43.3
GhostFaceNet	20.0	36.7	-	16.7	20.0
VGG-Face	56.7	56.7	43.3	20.0	-

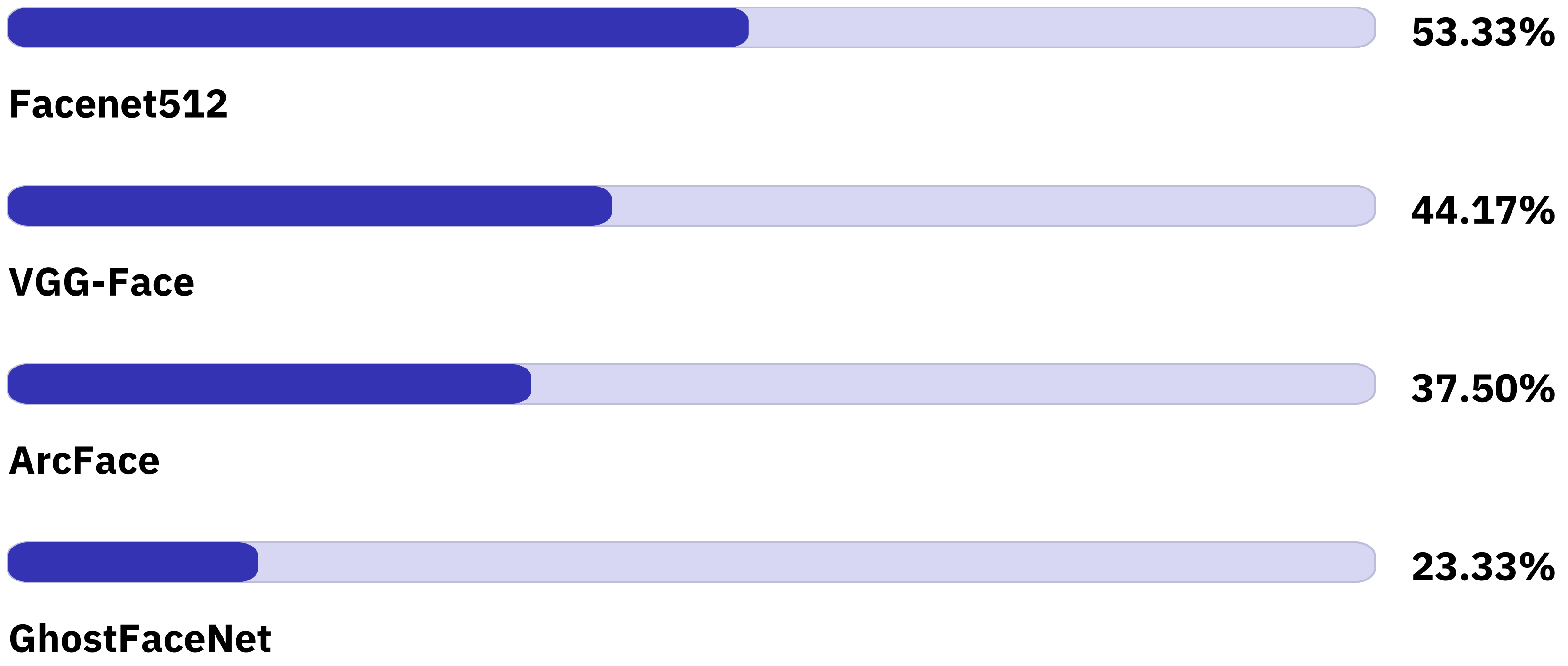
Top Transfer Pairs (Breach Rate): 1. Facenet512 → VGG-Face: 63.3%,
2. Facenet512 → ArcFace: 60.0%
3. VGG-Face → ArcFace / Facenet512: 56.7%

Weakest Transfer Pairs: GhostFaceNet → IR152: 16.7% | GhostFaceNet → Facenet512: 20.0% | GhostFaceNet → VGG-Face: 20.0% | VGG-Face → IR152: 20.0%

Mean Embedding Impact

Attacker ↓ / Victim →	Facenet512	ArcFace	GhostFaceNet	IR152	VGG-Face
Facenet512	-	0.244	0.144	0.288	0.231
ArcFace	0.316	-	0.148	0.235	0.162
GhostFaceNet	0.200	0.140	-	0.190	0.093
VGG-Face	0.451	0.230	0.137	0.208	-

Average Breach by Surrogate



Average Breach by Victim

Victim	Avg Breach
ArcFace	51.11%
GhostFaceNet	43.33%
VGG-Face	42.22%
Facenet512	37.78%
IR152	27.50%

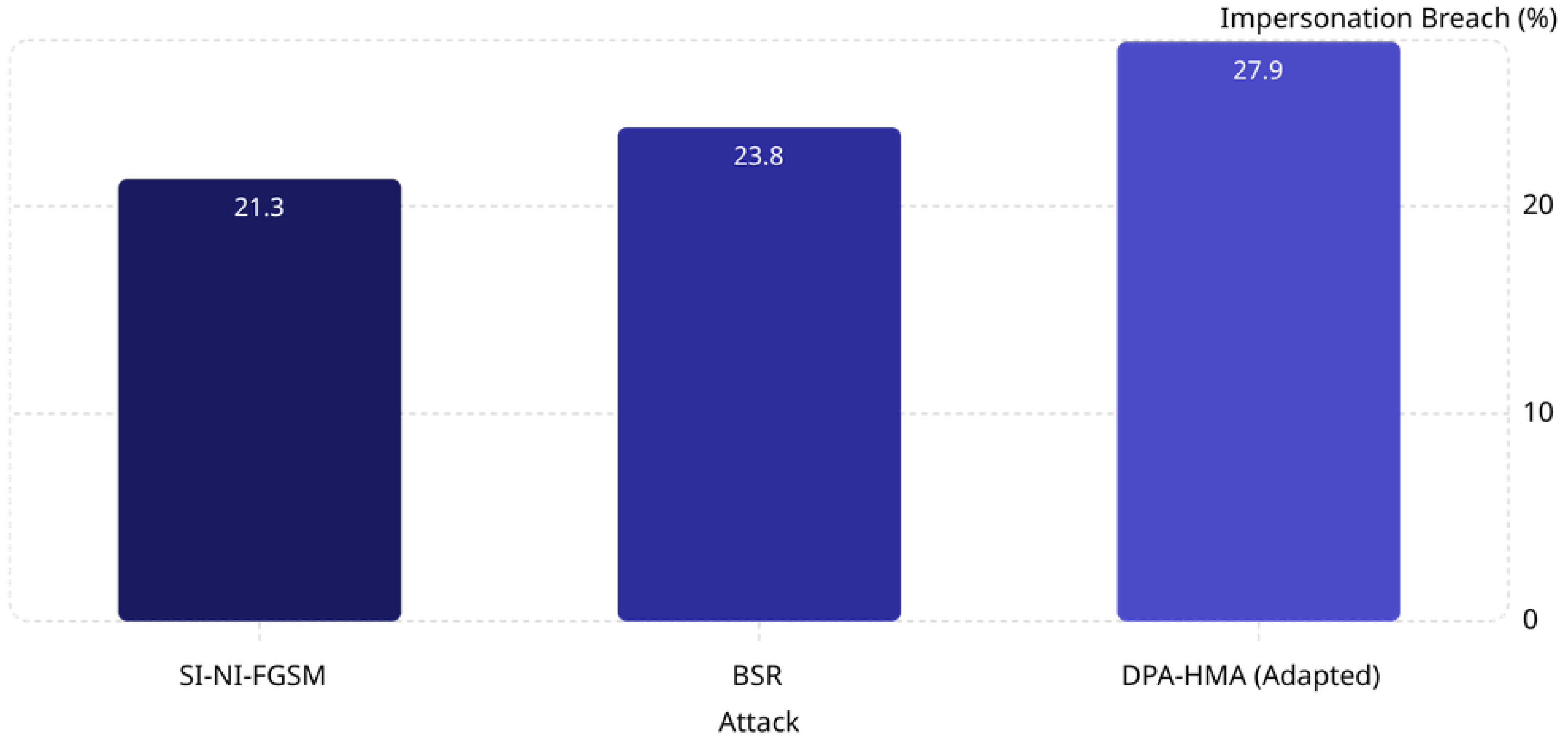
Comparison with BSR

Head-to-Head Cross-Model Breach Rate (%)

Attacker → Victim	DPA-HMA	BSR	Δ (pp)
Facenet512 → ArcFace	60.0	63.3	−3.3
Facenet512 → VGG-Face	63.3	56.7	**+6.6**
Facenet512 → IR152	46.7	50.0	−3.3
Facenet512 → GhostFaceNet	46.7	43.3	**+3.3**
VGG → Facenet512	56.7	60.0	−3.3
VGG → ArcFace	56.7	60.0	−3.3
ArcFace → Facenet512	36.7	36.7	0.0
Ghost → Facenet512	20.0	10.0	**+10.0**

- Overall: DPA-HMA **39.58%** vs BSR **36.5%** → **+3.1 pp**
- Impersonation: **+4.17 pp** (**27.92%** vs **23.75%**) - improved much on impersonation
- Dodging: DPA **51.25%** vs BSR **49.17%** - **+2.08 pp**
- Embedding Impact: **0.214** vs **0.205** - marginal difference

Impersonation Breach - Direct Comparison



DPA-HMA's aggregate impersonation advantage confirms that diverse-view gradient averaging improves the harder cross-identity transfer task, while also improving dodging over BSR.

Analysis - Mechanisms and Advantages

The performance gains of DPA-HMA (Adapted) are attributable to three synergistic mechanisms: gradient diversity, hard-view augmentation, and surrogate-dependent sensitivity.

Gradient Diversity (vs PGD / MI-FGSM)

- PGD: 16.7% - single gradient, no augmentation
- MI-FGSM: 26.7% - momentum but 1 view/step
- DPA-HMA: 8 warped copies/step → **+22.92 pp** over PGD, **+12.92 pp** over MI-FGSM
- Averaging over diverse views reduces surrogate-specific gradient overfitting

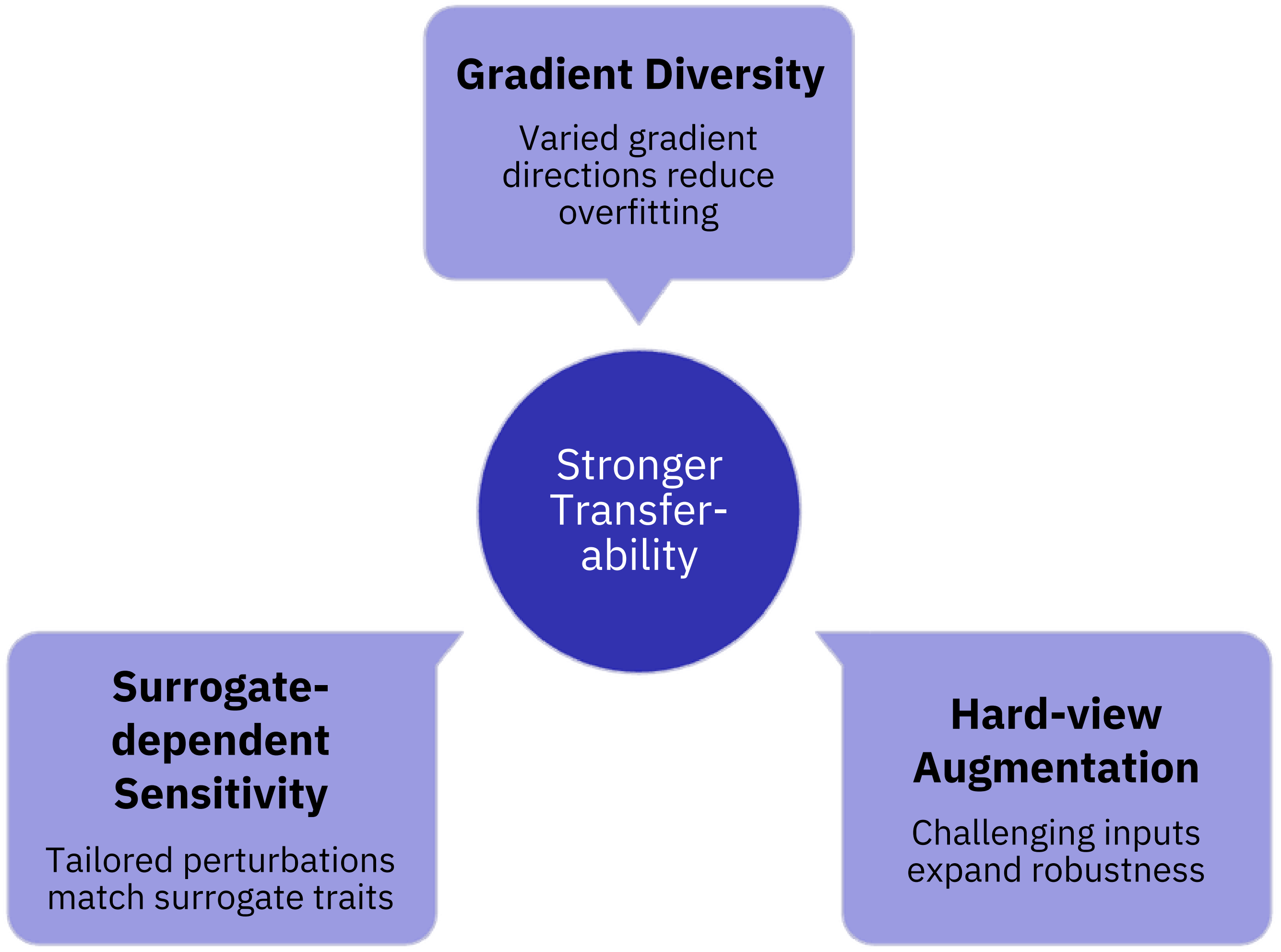
Hard-view + Geometric Augmentation (vs BSR)

- BSR uses block shuffle/rotate; DPA-HMA uses affine warps + hard pixel nudges
- Hard nudges explore local loss landscape (HMA-inspired)
- Affine warps model alignment and pose sensitivity in FR pipelines
- Result: **+4.17 pp impersonation** over BSR (27.92% vs 23.8%); Dodging: DPA 51.25% vs BSR 49.17% - **+2.08 pp**

Surrogate & Victim Sensitivity

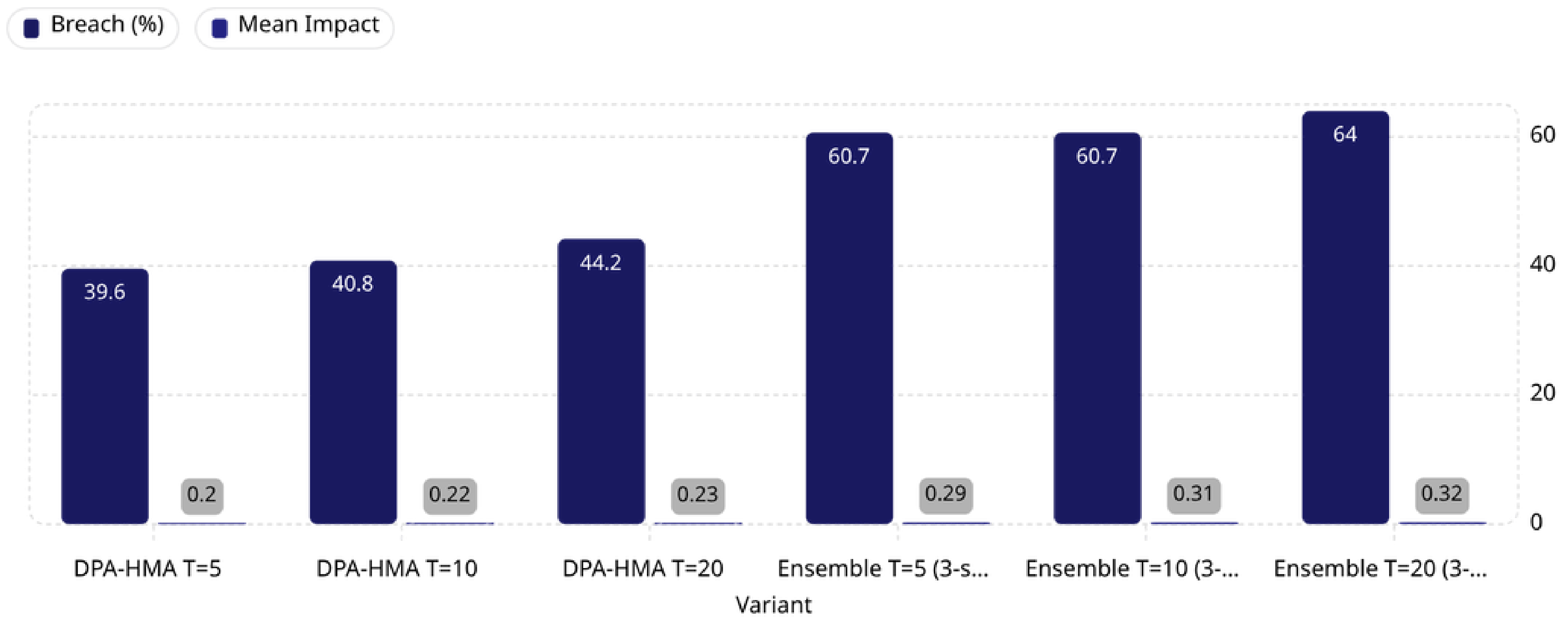
- Best surrogate: Facenet512 (53.33% avg cross-victim breach)
- Weakest surrogate: GhostFaceNet (23.3% avg cross victim) - architecture divergence limits transfer
- IR152 victim hardest (27.5% avg) - different PyTorch backbone vs DeepFace
- VGG-Face → Facenet512: impact 0.451 - strongest embedding shift among reported transfer pairs, even when breach rate is 56.7%.

Core Insight: Combining DPA + HMA yields +10.4 pp over the best vanilla and +3.1 pp over BSR - <u>with slightly higher mean embedding impact</u>



Ablation, Hyperparameters & Limitations

Ablation: Effect of Iteration Count & Ensemble



Ensemble Breach by Victim

Victim	Ensemble Breach (%)
ArcFace	70.0%
Facenet512	66.67%
GhostFaceNet	53.33%
VGG-Face	66.7%
IR152	63.33%

***Ensemble evaluated on 150 cases using 3-surrogate aggregation excluding self transfer cases. Attack Loss is computed for each surrogate and then averaged the three surrogate losses.**

e.g. Victim Model = Facenet512 & Surrogate Model = ArcFace + GhostFaceNet + VGG-Face

Hyperparameter Configuration

Parameter	Value
ε ($L\infty$ budget)	0.062
Iterations T	5 (submission)
Step size α	0.0124 (ε/T)
Momentum decay	1.0
Copies per step	8
Hard nudge steps	0, ±0.004, ±0.008
Affine noise σ	Uniform [0.01, 0.08]
Random seed	1 (reproducible)
Input normalization	pixels $\in [-1, 1]$
Canonical input size	224 × 224

Limitations

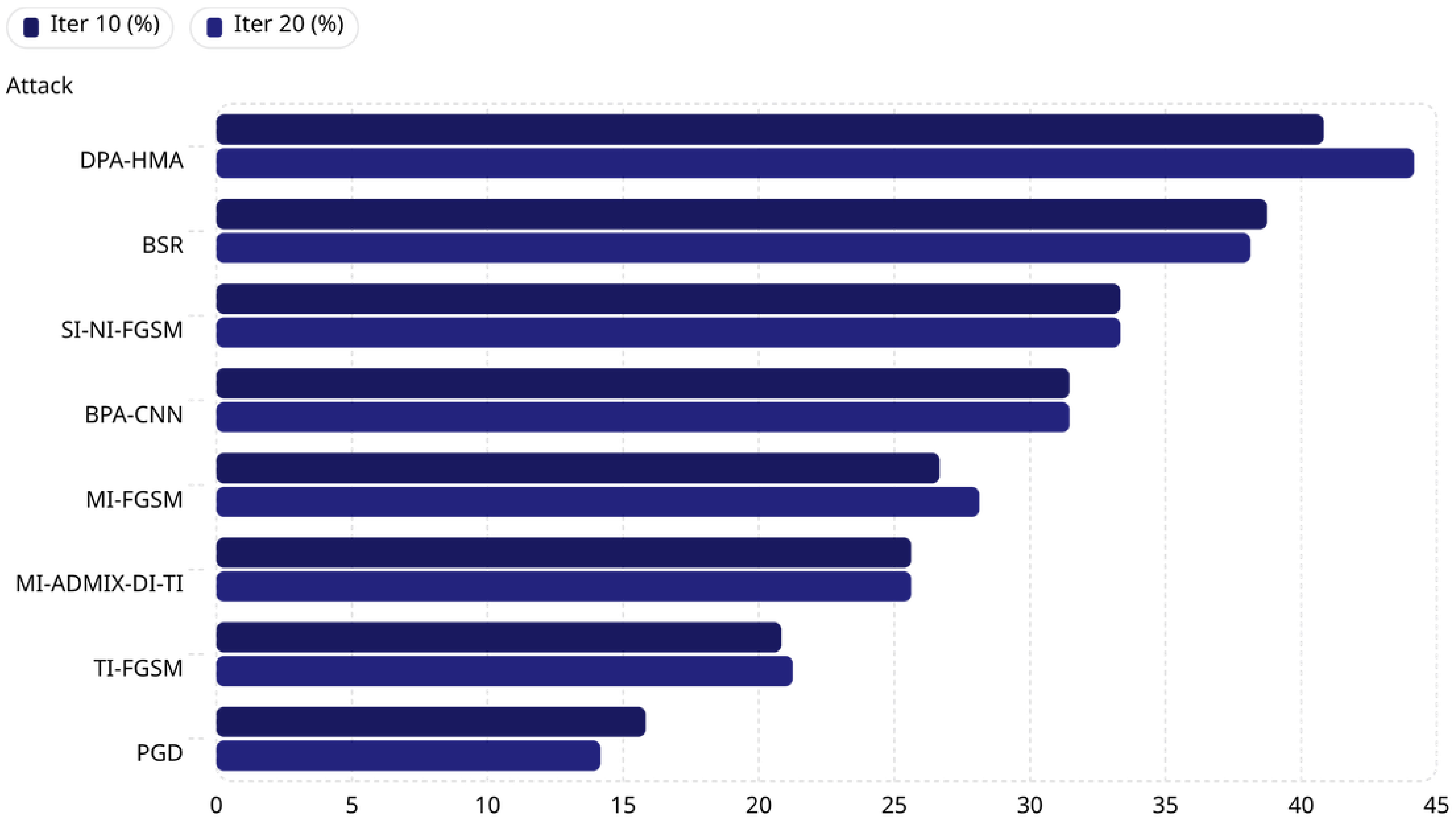
- Input-space adaptation only - not full paper multi-checkpoint or feature-map HMA
- GhostFaceNet surrogate: weak transfer (23.33% avg) due to architectural divergence
- VGGFace2 impersonation: 15.0% - high-diversity identity set challenge
- IT20 reaches 44.17% but requires 4x compute vs T=5

Reproducible at 39.58% breach / 0.214 impact with fixed random seed = 1.

DPA-HMA Gains More From Higher Iterations Than Standard Baselines

When all attacks are evaluated under the same 480-row protocol, increasing iterations from 10 to 20 gives DPA-HMA a clear boost, while standard baselines show only small changes or remain flat.

Iter 10 vs Iter 20 - Breach Rate Comparison



Full Comparison Table

Attack	Iter 10	Iter 20	Δ
DPA-HMA	**40.83%**	**44.17%**	**+3.34 pp**
BSR	38.75%	38.13%	−0.62 pp
SI-NI-FGSM	33.33%	33.33%	0.00 pp
BPA-CNN	31.46%	31.46%	0.00 pp
MI-FGSM	26.67%	28.13%	+1.46 pp
MI-ADMIX-DI-TI	25.63%	25.63%	0.00 pp
TI-FGSM	20.83%	21.25%	+0.42 pp
PGD	15.83%	14.17%	−1.66 pp

DPA-HMA: +3.34 pp (10→20 iters) largest gain. BSR regresses −0.62 pp. 4 of 8 attacks show zero gain.